Sahaj Bhadja

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Summary

Platform-focused Software Engineer with 2+ years in **Python** on Linux, building CI/CD and test automation, **Docker** and **Kubernetes** deployments, data and simulation pipelines, LLM workflows, and monitoring for reliable, cost-aware systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Restructured fault-tolerant **LLM workflow** orchestration with FastAPI background workers, explicit job states, and failure isolation, processing 1K+ AI jobs at 3x throughput without cascading failures under concurrent production load.
- Operated 5K+ monthly **data pipeline** jobs across PostgreSQL, SQL Server, and cloud storage with reconciliation checks and CI/CD-backed releases, sustaining 99%+ reliability in production for 500+ platform users.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Deployed a production backend on AWS through **GitHub Actions** CI/CD to Render and Vercel, serving 150+ instructors and 10K+ daily requests with stable responses during peak load.
- Tracked token usage and estimated **cost per model call** across GPT-4o, Claude, and Qwen-VL with automatic fallbacks, enabling cost-aware model use while cutting instructor feedback turnaround 60% in production.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built end-to-end **ML pipelines** processing 100K+ autonomous-driving trajectory logs with Hydra-configured training runs and Weights & Biases tracking, improving long-horizon trajectory planning 18% across closed-loop simulation runs.
- Ran **closed-loop simulation** evaluations comparing a map-aware encoder against three baselines on a single GPU, documenting failure modes and improving closed-loop planning stability 15% for autonomous vehicle behavior.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Automated nightly **CI pipelines** in Jenkins that allocated test VMs, built and installed executables on Linux, Windows, and macOS, ran 20+ suites, and emailed coverage reports and failure logs.
- Redesigned AWS Glue and S3 data pipelines, boosting efficiency 68% and cutting **storage costs** 15%, while optimizing APIs with caching and pagination to improve response times 30%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Provisioned infrastructure with **ARM templates** and packaged 8 microservices as Docker images through Jenkins CI/CD with staged, approval-gated, zero-downtime rolling releases to production, cutting deployment time 50% overall.
- Built **observability** with Sentry error tracking, Azure Monitor performance metrics, and Kibana log search, helping keep 2M+ synced records consistent across multiple locations in real time for operations teams.

Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, Bash, Shell
Platform: Linux, Docker, Kubernetes, ARM Templates, CI/CD, Jenkins, GitHub Actions, Git
Cloud: AWS (EC2, S3, RDS, Glue), Azure, GCP, Azure App Services, FastAPI, TCP/IP
Observability: Sentry, Azure Monitor, Kibana, Logging, Monitoring, Debugging, Troubleshooting, Root Cause Analysis, Postman
Data & AI: Data Pipelines, ETL, PostgreSQL, Spark, Databricks, LLM Systems, PyTorch, Hydra, Weights & Biases, Simulation

Projects

Simulation Job Platform on Kubernetes | *Kubernetes, Argo Workflows, Terraform, Prometheus, Grafana*

- Designed a **Kubernetes** platform that runs simulation and ML jobs as Argo Workflows with queue-based autoscaling, **Terraform**-managed clusters, and Grafana dashboards tracking job health and per-team compute cost.

Self-Hosted LLM Automation Service | *Python, vLLM, MQTT, Kong, OpenTelemetry*

- Designed an internal **self-hosted LLM** service behind an API gateway that consumes vehicle-log events from MQTT, drafts incident summaries with an agent, and exports traces through **OpenTelemetry**.

FinNexus Live | *Kubernetes, Next.js, TypeScript, MongoDB, CI/CD*

- Managed **Kubernetes** deployments and CI/CD automation for a fintech platform with Plaid bank linking and real-time updates, supporting 50,000+ users and 1M+ transactions with scalable, containerized services under high volume.